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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION APRIL 2021

BC303 - INTERMEDIARY METABOLISM-I

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer **ALL** questions

I. Choose the correct answer

1. According to IUB, enzymes are classified into---- classes
   a) 7 b) 4 c) 6 d) 8
2. The rate limiting enzyme in glycolysis is
   a) Phosphofructokinase b) Hexokinase
   c) Aldolase d) Phosphotriose isomerase
3. The coenzyme present in pyruvate carboxylase is
   a) Biotin b) TPP c) PLP d) Vitamin C
4. 2,4 dinitrophenol is an
   a) Uncoupler b) Inhibitor c) Ionophore d) All of the above
5. The net gain of ATP in glycolysis is
   a) 4 b) 2 c) 1 d) 6
6. Enzymes activity is controlled by
   a) pH of the solution b) Temperature
   c) Concentration of the enzyme d) Concentration of the substrate
7. In glycogenesis, Glucose should be converted to
   a) Glucuronic acid b) Pyruvic acid c) UDP glucose d) Sorbitol

8. Hexokinase has a high affinity for glucose than
   a) Fructokinase b) Galactokinase
   c) Glucokinase d) All of the above
9. The four membered aldose sugar phosphate formed in HMP shunt pathway is
   a) Xylulose P b) Erythrulose P c) Erythrose P d) Ribulose P
10. The respiratory chain complexes acting as proton pump are
    a) I, II and III b) I, II and IV c) I, III and IV d) I and II

II. True or False

11. TCA cycle operates in mitochondrial matrix.
12. Cyanide is an uncoupler of oxidative phosphorylation.
13. HMP shunt produces ATP molecules.
14. Transferases catalyzes the conversion of an aldose sugar to a ketose sugar
15. A specific inhibitor for succinate dehydrogenase is Malonate

III. Answer in One or Two Sentences

16. What is Allosteric site?
17. Define active site.
18. Pasteur Effect.
19. Write the significance of HMP shunt.
20. What is uncoupling agent?
21. What is a coenzyme?
22. How is competitive inhibition prevented?
23. How many ATPs are generated during anaerobic glycolysis?
24. What are glucogenic amino acids?

25. Which transport systemenergy conservation occurs?

SECTION-B (4X5=20)

Answer any **FOUR** questions

26. Write briefly on induced fit model of enzyme action.
27. Write brief notes on enzyme inhibition.
28. What are the components involved in ETC?
29. What are isoenzymes? Give examples.
30. What are the steps involved in glycogen metabolism?
31. Enumerate any one shuttle system.

SECTION-C (3X10=30)

Answer any **THREE** questions

32. Derive Michaelis-Menten's equation.
33. Give about the factors affecting the enzyme activity.
34. Explain the factors affecting the enzyme action.
35. State in detail about the oxidative and non-oxidative steps of HMP shunt pathway.
36. Describe in detail the Chemiosmotic theory

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